

Submission STAT1-MID-SUB01

Question STAT1-MID-Q01

scratch: show step

1. $\bar{x} = 25/5 = 5.00$

2. $s^2 = \sum(x_i - \bar{x})^2 / (5-1) = 14.00$

=> FINAL: mean=5.00; sample variance=14.00 (check units/interpretation)

Question STAT1-MID-Q02

scratch: show step

1. $P(D_1 \cap D_2) = (5/16)((5-1)/(16-1))$

2. =0.0833; the population shrinks after draw 1

=> FINAL: 0.0833 (check units/interpretation)

Question STAT1-MID-Q03

scratch: show step

1. $SE = \sqrt{p(1-p)/n} = \sqrt{0.55(1-0.55)/40} = 0.0787$

2. $CI = 0.55 \pm 1.96(0.0787) = (0.396, 0.704)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

=> FINAL: (0.396, 0.704); repeated-sampling interpretation (check units/interpretation)

Question STAT1-MID-Q04

scratch: show step

1. Increase sample size n to reduce standard error.

2. Reduce confidence level or population variability when justified.

3. Lower confidence changes coverage, so precision is not free.

=> FINAL: increase n; lower confidence level or variability; note the coverage trade-off (check units/interpretation)

Question STAT1-MID-Q05

scratch: show step

1. $0.03 < 0.05$, so reject H_0 under the stated rule.

2. Conclude evidence is inconsistent with H_0 ; do not say the probability H_0 is 3%.

=> FINAL: reject H_0 ; evidence against H_0 , not proof that H_0 is false (check units/interpretation)

Question STAT1-MID-Q06

scratch: show step

1. $\bar{x} = 35/5 = 7.00$

2. $s^2 = \sum(x_i - \bar{x})^2 / (5-1) = 4.50$

=> FINAL: mean=7.00; sample variance=4.50 (check units/interpretation)

Question STAT1-MID-Q07

scratch: show step

1. $P(D_1 \cap D_2) = (5/19)((5-1)/(19-1))$

2. $=0.0585$; the population shrinks after draw 1
=> FINAL: 0.0585 (check units/interpretation)

Question STAT1-MID-Q08

scratch: show step

1. $SE = \sqrt{p(1-p)/n} = \sqrt{0.35(1-0.35)/40} = 0.0754$

2. $CI = 0.35 \pm 1.96(0.0754) = (0.202, 0.498)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

=> FINAL: (0.202, 0.498); repeated-sampling interpretation (check units/interpretation)